

22050 Signals and Linear Systems in Continuous Time

Exam Formula Recipes

Contents

1	Systemklassifikation, LTIC, signaler, kausalitet	4
1.1	Ctrl+F keywords	4
1.2	Core formulas and recognition rules	4
1.3	Exam recipe: systemklassifikation, LTIC, lineært tidsinvariant kausalt	4
1.4	Exam recipe: multiple choice and one-best-answer elimination	4
1.5	Filter recognition, lavpas, højpas, bandpas	4
1.6	Signal power and energy, effekt	4
2	Impulsrespons, steprespons, ramprespons, zero-input	5
2.1	Ctrl+F keywords	5
2.2	Core relations	5
2.3	Direct feedthrough and impulse term	5
2.4	Exam recipe: impulsrespons from differential equation or overføringsfunktion	5
2.5	Exam recipe: steprespons to impulsrespons or ramprespons	5
2.6	Exam recipe: zero-input response, homogeneous response	5
2.7	Delayed input response, forsinket impuls	6
3	Foldning, convolution, graphical convolution	6
3.1	Ctrl+F keywords	6
3.2	Core formulas	6
3.3	Exam recipe: table convolution of causal exponentials	6
3.4	Exam recipe: graphical foldning	6
3.5	Impulse inside an integral	6
4	Fourier-række, Fourier series	6
4.1	Ctrl+F keywords	6
4.2	Core formulas	7
4.3	Symmetry checks	7
4.4	Exam recipe: Fourier-række statement questions	7
5	Fourier-transformation, Fourier transform, frequency response	7
5.1	Ctrl+F keywords	7
5.2	Definition and pairs	7
5.3	Properties used in exams	7
5.4	Symmetry and existence checks	8
5.5	Exam recipe: validate Fourier transform options	8
5.6	LTIC frequency response	8
6	Sampling, aliasing, modulation	8
6.1	Ctrl+F keywords	8

6.2	Sampling formulas	8
6.3	Aliasing and reconstruction	8
6.4	Exam recipe: sampling theorem or alias frequency	8
6.5	Modulation	8
7	ADC performance, quantization, ENOB	9
7.1	Ctrl+F keywords	9
7.2	Core formulas	9
7.3	Exam recipe: ADC numerical performance	9
8	Laplace, unilateral Laplace-transformation, partial fractions	9
8.1	Ctrl+F keywords	9
8.2	Definition and standard pairs	9
8.3	Unilateral rules with initial conditions	9
8.4	Initial and final value theorems	10
8.5	Exam recipe: inverse Laplace, step response, and brøkekspansion	10
8.6	Exam recipe: beginning value and final value	11
9	Overføringsfunktion, circuits, RLC, response calculation	11
9.1	Ctrl+F keywords	11
9.2	Transfer function from equation	11
9.3	Circuit relations and impedances	11
9.4	Exam recipe: differential equation from circuit	11
9.5	Exam recipe: transfer function and response	12
9.6	Poles, zeros, dimensions	12
9.7	Block diagram feedback transfer function	12
10	Second-order systems, Bode, damping, step metrics	12
10.1	Ctrl+F keywords	12
10.2	Second-order standard form	12
10.3	Second-order filter forms	12
10.4	Step response metrics	12
10.5	Bode rules	13
10.6	Exam recipe: identify Bode plot transfer function	13
11	Pole-zero filter design, transformations, tracking filters	13
11.1	Ctrl+F keywords	13
11.2	Pole-zero interpretation	13
11.3	Frequency transformations	13
11.4	Exam recipe: normalized lavpas to højpas	13
11.5	Tracking-filter asymptotic error	14
11.6	Exam recipe: zero steady-state error for step and ramp	14
12	Butterworth, Sallen-Key, sensitivitet	14
12.1	Ctrl+F keywords	14
12.2	Butterworth prototypes	14
12.3	Sallen-Key lowpass	14
12.4	Exam recipe: Sallen-Key node equations	14
12.5	Sensitivity, sensitivitet	15
12.6	Exam recipe: component sensitivity	15
12.7	Scaling	15

13 Butterworth højpas, AC coupling, instrumentation amplifier	15
13.1 Ctrl+F keywords	15
13.2 Butterworth highpass	15
13.3 AC coupling	15
13.4 Instrumentation amplifier	15
13.5 Exam recipe: AC-coupled in-amp checks	16
14 Exam tactics and edge cases	16
14.1 Ctrl+F keywords	16
14.2 One-best-answer and statement logic	16
14.3 Imaginary-axis poles and frequency response	16
14.4 Checklist before selecting an answer	16

1 Systemklassifkation, LTIC, signaler, kausalitet

1.1 Ctrl+F keywords

systemklassifkation, LTIC, lineært, linear, tidsinvariant, time invariant, kausalt, causal, stabil, BIBO, lige, ulige, even, odd, periodisk, impuls, step, rampe.

1.2 Core formulas and recognition rules

$$\begin{aligned}x_e(t) &= \frac{x(t) + x(-t)}{2}, & x_o(t) &= \frac{x(t) - x(-t)}{2}, & x(t + T_0) &= x(t), & \omega_0 &= \frac{2\pi}{T_0}. \\u(t) &= \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}, & r(t) &= tu(t), & \frac{d}{dt}u(t) &= \delta(t), & \int x(t)\delta(t - t_0) dt &= x(t_0). \\T\{ax_1 + bx_2\} &= aT\{x_1\} + bT\{x_2\}, & T\{x(t - t_0)\} &= y(t - t_0).\end{aligned}$$

Linearity	No $x^2, y^2, xy, \sin x, x $, products of dependent signals, or additive constants independent of input.
Time invariance	No explicit t -dependent coefficient such as $ty(t)$ or $\sin(t)$. Constant coefficients are OK.
Causality	No future input $x(t + a)$, $a > 0$. Differential equations with present input and finite derivatives are causal in these exams.
LTIC form	$\sum a_k D^k y(t) = \sum b_m D^m x(t)$, real constant coefficients.
Stability quick test	For rational causal LTIC: all poles strictly in left half-plane. Imaginary-axis poles: marginal/not BIBO stable.

1.3 Exam recipe: systemklassifkation, LTIC, lineært tidsinvariant kausalt

Use when: asked which system is LTIC, linear, time-invariant, or causal.

1. Check linearity first: reject powers/products/nonlinear functions of x, y and constant offsets.
2. Check time invariance: reject explicit t -dependent coefficients or independent sources like $\sin t$.
3. Check causality: reject future input $x(t + a)$, $a > 0$.
4. If an option asks for LTIC, all three tests must pass.

Fast checks: $a_0 y + \dot{y} = b_0 x$ is LTIC; $\dot{y} + ty = x$, $\dot{y} + y = x^2$, $\dot{y} + y = x(t + 2)$ fail. **Common traps:** additive $+1$ or $\sin t$ destroys linearity even if the equation looks first order.

1.4 Exam recipe: multiple choice and one-best-answer elimination

Use when: options contain several claims or the exam says one-best-answer.

1. Mark every clause in an option true/false.
2. One false clause makes the whole option false.
3. If the exam has a “more than one true option” choice, use it only when at least two complete options are fully true.
4. For “not correct” questions, invert the final selection only after checking all clauses.

Fast checks: test limiting cases, dimensions, signs, pole locations, and DC/high-frequency gain before doing long algebra.

1.5 Filter recognition, lavpas, højpas, bandpas

$$\begin{aligned}H(0) \neq 0, H(j\infty) = 0 &\Rightarrow \text{lavpas/lowpass}, & H(0) = 0, H(j\infty) \neq 0 &\Rightarrow \text{højpas/highpass}. \\H(0) = 0, H(j\infty) = 0 &\Rightarrow \text{bandpas/bandpass}, & H(j\omega_0) = 0 &\Rightarrow \text{notch at } \omega_0.\end{aligned}$$

Trap: count powers of s in numerator and denominator, not visual complexity.

1.6 Signal power and energy, effekt

$$\begin{aligned}E_x &= \int_{-\infty}^{\infty} |x(t)|^2 dt, & P_x &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt. \\& & \frac{1}{T_0} \int_{T_0} & |A \cos(\omega_0 t + \phi)|^2 dt = \frac{A^2}{2}.\end{aligned}$$

Use when: asked for signalets effekt/power of a periodic sinusoid. **Trap:** power is average squared magnitude, not amplitude.

2 Impulsrespons, steprespons, ramprespons, zero-input

2.1 Ctrl+F keywords

impulsrespons, impulse response, enhedsimpuls, steprespons, enhedstrinrespons, unit step response, ramprespons, enhedsramperrespons, direct feedthrough, direkte impulsled, zero-input, zero-state, begyndelsesværdi.

2.2 Core relations

$$Q(D)y(t) = P(D)x(t), \quad H(s) = \frac{Y(s)}{X(s)} = \frac{P(s)}{Q(s)} \quad \text{for zero initial conditions.}$$

$$h(t) = \mathcal{L}^{-1}\{H(s)\}, \quad y_{zs}(t) = h(t) * x(t), \quad y(t) = y_{zi}(t) + y_{zs}(t).$$

$$y_{step}(t) = h(t) * u(t) = \int_{0^-}^t h(\tau) d\tau, \quad h(t) = \frac{d}{dt} y_{step}(t).$$

$$y_{ramp}(t) = h(t) * r(t) = \int_{0^-}^t y_{step}(\tau) d\tau, \quad Y_{step}(s) = \frac{H(s)}{s}, \quad Y_{ramp}(s) = \frac{H(s)}{s^2}.$$

2.3 Direct feedthrough and impulse term

If $H(s) = k + H_{strict}(s)$, then

$$h(t) = k\delta(t) + \mathcal{L}^{-1}\{H_{strict}(s)\}.$$

For $H(s) = \frac{b_n s^n + \dots}{s^n + a_{n-1} s^{n-1} + \dots}$, the direct impulse coefficient is b_n . If numerator order is lower than denominator order, there is no $\delta(t)$ term.

2.4 Exam recipe: impulsrespons from differential equation or overføringsfunktion

Use when: asked “Hvad er impulsresponsen?” or response to $x(t) = \delta(t)$.

1. Set initial conditions to zero unless zero-input is explicitly requested.
2. Form $H(s) = P(s)/Q(s)$.
3. Divide if numerator degree equals denominator degree; keep the quotient as $k\delta(t)$.
4. Partial-fraction the strictly proper remainder.
5. Invert with causal $u(t)$ factors.

Fast checks: $h(0^+) = \lim_{s \rightarrow \infty} sH_{strict}(s)$ only after removing any $\delta(t)$ term. **Common traps:** beginning value theorem does not return the strength of a $\delta(t)$ term.

2.5 Exam recipe: steprespons to impulsrespons or ramprespons

Use when: given y_{step} and asked about $h(t)$ or $y_{ramp}(t)$.

1. Differentiate y_{step} to get $h(t)$; remember jumps create impulses.
2. Integrate y_{step} from 0^- to t to get y_{ramp} .
3. Keep the causal factor $u(t)$ on final expressions.
4. Check final value: $y_{step}(\infty) = H(0)$ if stable.

Trap: differentiating $au(t)$ gives $a\delta(t)$.

2.6 Exam recipe: zero-input response, homogeneous response

Use when: initial conditions $y(0^-), \dot{y}(0^-)$ are nonzero or exam asks zero-input.

1. Set $x(t) = 0$.
2. Solve $Q(D)y_{zi} = 0$ from characteristic roots.
3. Use unilateral derivative rules to include 0^- values if solving in s -domain.
4. Stable systems have $y_{zi}(t) \rightarrow 0$; unstable poles do not.

Fast checks: zero-input depends only on initial conditions and $Q(D)$, not on $P(D)$.

2.7 Delayed input response, forsinket impuls

$$x(t - t_0)u(t - t_0) \Rightarrow y_{zs}(t - t_0)u(t - t_0), \quad \delta(t - t_0) \Rightarrow h(t - t_0).$$

Use when: asked response to delayed input or delayed impulse. First solve the undelayed response, then shift the answer by t_0 . **Trap:** do not change the shape or coefficients when only the input is delayed in an LTIC system.

3 Foldning, convolution, graphical convolution

3.1 Ctrl+F keywords

foldning, foldningsintegral, convolution, $f * g$, shifted step, forskudt stepfunktion, rampe, grafisk foldning, impulse sampling.

3.2 Core formulas

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau.$$

$$\delta(t - t_0) * f(t) = f(t - t_0), \quad u(t) * u(t) = r(t) = tu(t).$$

$$e^{\lambda_1 t}u(t) * e^{\lambda_2 t}u(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}u(t), \quad \lambda_1 \neq \lambda_2.$$

$$e^{\lambda t}u(t) * e^{\lambda t}u(t) = te^{\lambda t}u(t), \quad e^{\lambda t}u(t) * te^{\lambda t}u(t) = \frac{1}{2}t^2e^{\lambda t}u(t).$$

$$u(t) * r(t) = \frac{1}{2}t^2u(t), \quad u(t - t_0) * r(t) = \frac{1}{2}(t - t_0)^2u(t - t_0).$$

3.3 Exam recipe: table convolution of causal exponentials

Use when: both factors look like $Ae^{-at}u(t)$, $Bte^{-bt}u(t)$, $u(t)$, or $r(t)$.

1. Pull constants outside the convolution.
2. Match the pair to the table above or multiply Laplace transforms.
3. If exponents are equal, use the repeated-pole result $te^{\lambda t}u(t)$.
4. Apply any delay last: $f(t - t_0)u(t - t_0)$.

Fast checks: causal inputs produce causal output; output starts at the sum of start times. **Common traps:** $\lambda_1 - \lambda_2$ sign; forgetting the final $u(t)$.

3.4 Exam recipe: graphical foldning

Use when: asked which graph represents $f * g$.

1. Find first and last possible overlap: support endpoints add.
2. The convolution is zero outside the overlap interval.
3. Area of overlap determines sign and shape; rectangular overlap gives ramps/triangles.
4. Use continuity: convolution of ordinary bounded signals is usually continuous at breakpoints.

Fast checks: widths add; peak often occurs where overlap area is largest.

3.5 Impulse inside an integral

$$\int_{-\infty}^{\infty} \delta(\tau - a)f(t - \tau) d\tau = f(t - a), \quad \int \delta(t - \tau - a)f(\tau) d\tau = f(t - a).$$

Trap: sample the variable of integration, not the free variable.

4 Fourier-række, Fourier series

4.1 Ctrl+F keywords

Fourier-række, Fourier series, eksponentiel Fourier-række, harmonisk, koefficienter, D_n , C_n , real even, reel lige, odd, ulige, periodic extension.

4.2 Core formulas

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, \quad D_n = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jn\omega_0 t} dt.$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t),$$

$$a_0 = D_0, \quad a_n = D_n + D_{-n}, \quad b_n = j(D_n - D_{-n}).$$

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n), \quad C_n = 2|D_n|, \quad \theta_n = \arg(D_n).$$

4.3 Symmetry checks

Real signal	$D_{-n} = D_n^*$.
Real and even, reel lige	D_n real and even in n ; sine terms vanish.
Real and odd, reel ulige	D_n imaginary and odd in n ; cosine terms vanish.
Half-wave symmetry	even harmonics often vanish.

$$\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |D_n|^2.$$

4.4 Exam recipe: Fourier-række statement questions

Use when: asked which statement about exponential Fourier series is true.

1. Check if the signal is periodic; Fourier series represents periodic signals.
2. For an aperiodic time window of length T_0 , the reconstructed signal is the periodic extension with period T_0 .
3. Match coefficient symmetry to signal symmetry.
4. Check basis function form: $D_n e^{jn\omega_0 t}$.

Common traps: real even signals do not have imaginary odd coefficients; that is the real odd case.

5 Fourier-transformation, Fourier transform, frequency response

5.1 Ctrl+F keywords

Fourier, Fouriertransformation, Fourier transform, fourierdomænet, frekvenskarakteristik, magnitude, phase, spectrum, sinc, symmetry, modulation, time scaling.

5.2 Definition and pairs

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt, \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

$$\delta(t) \leftrightarrow 1, \quad 1 \leftrightarrow 2\pi\delta(\omega), \quad e^{-at}u(t) \leftrightarrow \frac{1}{a + j\omega} \quad (a > 0).$$

$$\cos(\omega_0 t) \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)],$$

$$\text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right), \quad \text{sinc}(x) = \frac{\sin x}{x}.$$

5.3 Properties used in exams

$$\frac{dx}{dt} \leftrightarrow j\omega X(\omega), \quad x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega).$$

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right), \quad x(t) e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0).$$

$$x(t) * h(t) \leftrightarrow X(\omega) H(\omega), \quad x(t) h(t) \leftrightarrow \frac{1}{2\pi} X(\omega) * H(\omega).$$

5.4 Symmetry and existence checks

Real $x(t)$	$X(-\omega) = X^*(\omega)$; magnitude even, phase odd.
Real even $x(t)$	$X(\omega)$ real and even.
Real odd $x(t)$	$X(\omega)$ imaginary and odd.
Imaginary odd $x(t)$	$X(\omega)$ real and even.
Periodic signal	Ordinary transform uses impulses; Fourier series often easier.

5.5 Exam recipe: validate Fourier transform options

Use when: options modify a known pair $x(t) \leftrightarrow X(\omega)$.

1. For derivative, multiply by $j\omega$.
2. For $x(at)$, scale by $1/|a|$ and replace ω by ω/a .
3. For $x(t)e^{j\omega_0 t}$, shift spectrum to $X(\omega - \omega_0)$.
4. Use symmetry to reject impossible real/imaginary claims.

Common traps: missing $1/|a|$; wrong sign in modulation shift.

5.6 LTIC frequency response

$$Y(\omega) = H(\omega)X(\omega), \quad H(\omega) = H(s)|_{s=j\omega}$$

when the ordinary frequency response exists. Imaginary-axis poles require care; substitution may be singular.

6 Sampling, aliasing, modulation

6.1 Ctrl+F keywords

sampling, sample, Nyquist, aliasing, rekonstruktion, impulse train, amplitudemodulation, modulation, spectrum copies.

6.2 Sampling formulas

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s), \quad f_s = \frac{1}{T_s}, \quad \omega_s = \frac{2\pi}{T_s}.$$

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

$$\omega_s > 2\omega_{\max} \Leftrightarrow f_s > 2f_{\max}$$

for no aliasing of a bandlimited baseband signal.

6.3 Aliasing and reconstruction

$$\omega_{\text{alias}} = |\omega_0 - k\omega_s| \quad \text{chosen in } [0, \omega_s/2].$$

$$X(\omega) = T_s X_s(\omega) \quad \text{inside the reconstruction lowpass passband.}$$

6.4 Exam recipe: sampling theorem or alias frequency

Use when: asked whether sampling is sufficient or where a sinusoid aliases.

1. Convert all frequencies to the same unit: Hz or rad/s.
2. Compare f_s with $2f_{\max}$.
3. For a sinusoid above Nyquist, subtract integer multiples of f_s until inside $[0, f_s/2]$.
4. Check whether spectrum copies overlap.

Common traps: mixing f and $\omega = 2\pi f$; equality at Nyquist is not a robust no-aliasing condition.

6.5 Modulation

$$x(t) \cos(\omega_c t) \leftrightarrow \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)].$$

Fast check: multiplying by a cosine creates two half-amplitude shifted spectra.

7 ADC performance, quantization, ENOB

7.1 Ctrl+F keywords

ADC, quantization, kvantisering, LSB, SQNR, SNR, SINAD, ENOB, THD, SFDR, jitter, oversampling.

7.2 Core formulas

$$\begin{aligned}\Delta &= \frac{V_{\text{FS}}}{2^N}, & e_q &\in [-\Delta/2, \Delta/2], & P_q &= \frac{\Delta^2}{12}. \\ \text{SQNR}_{\text{dB}} &\approx 6.02N + 1.76 & \text{for full-scale sine.} \\ \text{ENOB} &= \frac{\text{SINAD}_{\text{dB}} - 1.76}{6.02}, & \text{THD}_{\text{dB}} &= 10 \log_{10} \frac{\sum_{k \geq 2} P_k}{P_1}. \\ \text{SFDR}_{\text{dB}} &= 10 \log_{10} \frac{P_{\text{carrier}}}{P_{\text{largest spur}}}, & \text{SNR}_{j,\text{dB}} &\approx -20 \log_{10}(2\pi f_{\text{in}} \sigma_t). \\ \text{Process gain} &= 10 \log_{10}(\text{OSR}).\end{aligned}$$

7.3 Exam recipe: ADC numerical performance

Use when: asked for resolution, noise, ENOB, SINAD, THD, SFDR, or jitter limit.

1. Convert powers to dB with $10 \log_{10}$; amplitudes with $20 \log_{10}$.
2. Use $6.02N + 1.76$ only for ideal full-scale sine SQNR.
3. Use SINAD, not SNR, when computing ENOB.
4. For jitter, check the input frequency; jitter noise increases with f_{in} .

Common traps: using full-scale peak instead of full-scale range; mixing amplitude and power dB.

8 Laplace, unilateral Laplace-transformation, partial fractions

8.1 Ctrl+F keywords

Laplace, unilateral, laplacetransformation, inverse Laplace, partial fraction, brøkekspansion.

Begyndelsesværdi, slutværdi, beginning value theorem, final value theorem, ROC, konvergensområde.

8.2 Definition and standard pairs

$$\begin{aligned}X(s) &= \mathcal{L}\{x(t)\} = \int_{0^-}^{\infty} x(t)e^{-st} dt. \\ \delta(t) &\leftrightarrow 1, & u(t) &\leftrightarrow \frac{1}{s}, & r(t) = tu(t) &\leftrightarrow \frac{1}{s^2}. \\ e^{-at}u(t) &\leftrightarrow \frac{1}{s+a}, & te^{-at}u(t) &\leftrightarrow \frac{1}{(s+a)^2}. \\ e^{-\alpha t} \cos \omega_d t u(t) &\leftrightarrow \frac{s+\alpha}{(s+\alpha)^2 + \omega_d^2}, \\ e^{-\alpha t} \sin \omega_d t u(t) &\leftrightarrow \frac{\omega_d}{(s+\alpha)^2 + \omega_d^2}.\end{aligned}$$

8.3 Unilateral rules with initial conditions

$$\begin{aligned}\mathcal{L}\{\dot{x}\} &= sX(s) - x(0^-), & \mathcal{L}\{\ddot{x}\} &= s^2X(s) - sx(0^-) - \dot{x}(0^-). \\ \int_{0^-}^t x(\tau) d\tau &\leftrightarrow \frac{X(s)}{s}, & x(t-t_0)u(t-t_0) &\leftrightarrow e^{-st_0}X(s). \\ x * h &\leftrightarrow X(s)H(s), & \text{causal rational ROC: } \text{Re } s > \text{Re}(p_{\text{max}}).\end{aligned}$$

8.4 Initial and final value theorems

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s)$$

if the ordinary right limit exists and impulses at $t = 0$ have been handled.

$$x(\infty) = \lim_{s \rightarrow 0} sX(s)$$

if all poles of $sX(s)$ are strictly in the left half-plane.

8.5 Exam recipe: inverse Laplace, step response, and brøkekspansion

Use when: asked for $y(t)$, $h(t)$, step response, impulse response, or when matching a Laplace expression to answer options.

8.5.1 Step response and impulse response recognition

$$Y_{\text{step}}(s) = H(s) \frac{1}{s}, \quad Y_{\text{impulse}}(s) = H(s).$$

Step response adds one pole at $s = 0$. Impulse response is the inverse Laplace transform of $H(s)$.

8.5.2 Final value and DC gain

$$y(\infty) = \lim_{s \rightarrow 0} sY(s), \quad y_{\text{step}}(\infty) = H(0).$$

Use the final value theorem only when the relevant poles are stable. In MCQ, check the final value before doing full partial fractions.

8.5.3 Pole terms in inverse Laplace

$$\begin{aligned} s + a &\Rightarrow e^{-at}. \\ s^2 + 2\alpha s + (\alpha^2 + \omega_d^2) &= (s + \alpha)^2 + \omega_d^2. \\ e^{-\alpha t} \cos(\omega_d t), & \quad e^{-\alpha t} \sin(\omega_d t). \end{aligned}$$

Real poles give exponential terms. Complex conjugate poles give damped sinus and cosinus terms.

8.5.4 Multiple-choice elimination checks

- final value
- exponential decay/growth
- sinus/cosinus frequency
- initial value $y(0^+)$, if useful
- sign of the first movement, if useful

Only do full brøkekspansion if several answer options survive.

8.5.5 Quadratic factor brøkekspansion

$$\begin{aligned} \frac{1}{s(s^2 + 2\alpha s + \omega_n^2)} &= \frac{A}{s} + \frac{Bs + C}{s^2 + 2\alpha s + \omega_n^2}. \\ A &= sY(s)|_{s=0}. \\ Bs + C &= B(s + \alpha) + (C - B\alpha). \end{aligned}$$

Find the constant step term A first. Rewrite the numerator using $s + \alpha$ before matching cosine and sine transform pairs.

8.5.6 Damped sinusoid transform matching

$$\begin{aligned} \frac{s + \alpha}{(s + \alpha)^2 + \omega_d^2} &\leftrightarrow e^{-\alpha t} \cos(\omega_d t), \\ \frac{\omega_d}{(s + \alpha)^2 + \omega_d^2} &\leftrightarrow e^{-\alpha t} \sin(\omega_d t). \end{aligned}$$

The real pole part sets $e^{-\alpha t}$. The imaginary pole part sets ω_d .

8.5.7 Causal inverse Laplace transform

$$x(t) = \mathcal{L}^{-1}\{X(s)\}u(t).$$

Add $u(t)$ for causal inverse Laplace transforms.

8.5.8 Fast checks for step response and brøkekspansion

- Stable step response final value: $H(0)$.
- Complex poles: real part gives $e^{-\alpha t}$.
- Imaginary part gives $\cos(\omega_d t)$ and $\sin(\omega_d t)$.
- Denominator constant in $s^2 + 2\zeta\omega_n s + \omega_n^2$ gives ω_n^2 , not ω_n .
- Strictly proper systems with zero initial conditions and step input often have $y(0^+) = 0$.

8.6 Exam recipe: beginning value and final value

Use when: asked $h(0^+)$, $y(0^+)$, asymptotic value, or steady-state error.

1. For 0^+ , remove direct $\delta(t)$ terms before using $\lim_{s \rightarrow \infty} sX(s)$.
2. For ∞ , check poles of $sX(s)$; any right-half-plane or nonzero imaginary-axis pole invalidates FVT.
3. Evaluate the limit only after the conditions pass.

Common traps: FVT cannot prove a limit for unstable or oscillatory signals.

9 Overføringsfunktion, circuits, RLC, response calculation

9.1 Ctrl+F keywords

overføringsfunktion, transfer function, differentiallyigning fra kredsløb, RLC, knudepunkt, nodal equation, voltage divider, RC lavpas, RC højpas, zero-state, zero-input.

9.2 Transfer function from equation

$$\sum_{k=0}^N a_k D^k y(t) = \sum_{m=0}^M b_m D^m x(t) \Rightarrow H(s) = \frac{\sum_{m=0}^M b_m s^m}{\sum_{k=0}^N a_k s^k}$$

with zero initial conditions.

$$H(j\omega) = H(s)|_{s=j\omega} \quad \text{when no singularity blocks the frequency response.}$$

9.3 Circuit relations and impedances

$$v_R = Ri_R, \quad i_C = C \frac{dv_C}{dt}, \quad v_L = L \frac{di_L}{dt}.$$

$$Z_R = R, \quad Z_C = \frac{1}{sC}, \quad Z_L = sL.$$

$$H(s) = \frac{Z_{\text{out}}}{Z_1 + Z_2} \quad \text{for a simple unloaded divider.}$$

$$H_{RC,LP}(s) = \frac{1}{1 + sRC}, \quad H_{RC,HP}(s) = \frac{sRC}{1 + sRC}.$$

9.4 Exam recipe: differential equation from circuit

Use when: asked “differentiallyigning fra kredsløb” or RLC component laws.

1. Label node voltages; choose $x(t)$ and $y(t)$ exactly as in the problem.
2. Write KCL in time domain or s -domain.
3. Use $i_C = C\dot{v}_C$, $v_L = L\dot{i}_L$, $v_R = Ri_R$.
4. Rearrange to all y -terms on left and x -terms on right.
5. Identify whether numerator has x , $\dot{x}$, or $\ddot{x}$ terms.

Fast checks: a series capacitor at input often creates highpass behavior; output across capacitor often lowpass.

9.5 Exam recipe: transfer function and response

Use when: asked $H(s)$, $Y(s)$, or response to $u(t)$, $r(t)$, $e^{-at}u(t)$.

1. Find $H(s)$ with zero initial conditions.
2. Multiply by $X(s)$: $1/s$ for step, $1/s^2$ for ramp, $1/(s+a)$ for $e^{-at}u(t)$.
3. Add zero-input terms separately if initial conditions are nonzero.
4. Partial-fraction and invert.

Common traps: unilateral initial-condition terms are not part of the transfer function.

9.6 Poles, zeros, dimensions

$$H(s) = K \frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}, \quad \text{zeros: numerator roots, poles: denominator roots.}$$

Dimension check: if $Y(s) = H(s)X(s)$, $H(s)$ is dimensionless for voltage gain. $h(t)$ has dimension $1/s$ when convolved with $x(t) dt$ to produce $y(t)$.

9.7 Block diagram feedback transfer function

$$T(s) = \frac{Y(s)}{X(s)} = \frac{G(s)}{1 + G(s)H_f(s)}$$

for negative feedback with forward path G and feedback path H_f .

$$T(s) = \frac{G(s)}{1 - G(s)H_f(s)}$$

for positive feedback. **Fast checks:** summing-junction sign decides $+$ or $-$ in the denominator; open-loop blocks multiply along a path.

10 Second-order systems, Bode, damping, step metrics

10.1 Ctrl+F keywords

2. ordens, second order, dæmpningsfaktor, damping, ζ , ω_n , underdæmpet, overdæmpet, kritisk dæmpet, Bode, resonance, overshoot, settling time.

10.2 Second-order standard form

$$s^2 + 2\zeta\omega_n s + \omega_n^2, \quad p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}.$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}, \quad Q = \frac{1}{2\zeta}.$$

$0 < \zeta < 1$	underdamped, complex conjugate poles, ringing.
$\zeta = 1$	critically damped, repeated real pole.
$\zeta > 1$	overdamped, two real negative poles.
$\zeta \leq 0$	unstable or marginal depending on poles.

10.3 Second-order filter forms

$$H_{LP}(s) = K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad H_{HP}(s) = K \frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

$$H_{BP}(s) = K \frac{2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

10.4 Step response metrics

For $0 < \zeta < 1$:

$$t_p = \frac{\pi}{\omega_d}, \quad PO = 100e^{-\zeta\pi/\sqrt{1-\zeta^2}}\%, \quad t_s \approx \frac{4}{\zeta\omega_n} \text{ (2 percent).}$$

$$\omega_r = \omega_n\sqrt{1-2\zeta^2} \quad \text{only if } \zeta < \frac{1}{\sqrt{2}}.$$

10.5 Bode rules

$$|H|_{\text{dB}} = 20 \log_{10} |H(j\omega)|, \quad \angle H = \arg H(j\omega).$$

Constant K	$20 \log_{10} K $, phase 0° or 180° .
Zero at origin s	$+20$ dB/dec, $+90^\circ$.
Pole at origin $1/s$	-20 dB/dec, -90° .
Real zero $1 + s/\omega_z$	slope increases $+20$ dB/dec after ω_z .
Real pole $1/(1 + s/\omega_p)$	slope decreases -20 dB/dec after ω_p .
Second-order poles	slope decreases -40 dB/dec after ω_n , phase about -180° .

10.6 Exam recipe: identify Bode plot transfer function

Use when: asked which $H(s)$ matches a Bode plot.

1. Read low-frequency slope: zeros/poles at origin.
2. Read high-frequency slope: relative numerator/denominator degree.
3. Locate break frequency or ω_n .
4. Use phase end values to distinguish numerator s^2 , denominator s^2 , and sign errors.
5. Reject unstable denominators with negative damping unless the plot explicitly implies instability.

Fast checks: highpass starts low and rises; lowpass starts flat and falls.

11 Pole-zero filter design, transformations, tracking filters

11.1 Ctrl+F keywords

pole-zero, poler, nulpunkter, filterdesign, lavpas, højpas, bandpas, notch, knækfrekvens, frequency transformation, tracking-filter, asymptotisk fejl.

11.2 Pole-zero interpretation

$$H(s) = K \frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}, \quad |H(j\omega)| = |K| \frac{\prod_i |j\omega - z_i|}{\prod_k |j\omega - p_k|}.$$

Zeros at $s = 0$: highpass behavior. Poles near the imaginary axis: resonance/large peak. Zeros at $\pm j\omega_0$: notch at ω_0 .

11.3 Frequency transformations

Lowpass prototype $H_{LP}(s)$:

$$\text{frequency scaling to cutoff } \omega_c : \quad H(s) = H_{LP}\left(\frac{s}{\omega_c}\right).$$

$$\text{lavpas to højpas/highpass cutoff } \omega_c : \quad H_{HP}(s) = H_{LP}\left(\frac{\omega_c}{s}\right).$$

$$\text{lavpas to bandpas: } s \mapsto \frac{s^2 + \omega_0^2}{Bs}.$$

11.4 Exam recipe: normalized lavpas to højpas

Use when: question asks which substitution gives a highpass with ω_n or cutoff.

1. Start from normalized lowpass denominator in variable s .
2. Substitute $s \leftarrow \omega_c/s$ or $s \leftarrow \omega_n/s$ as specified.
3. Multiply numerator and denominator by powers of s to make a polynomial.
4. Check $H(0) = 0$ and $H(\infty) \neq 0$.

Common traps: s/ω_c only frequency-scales lowpass; it does not make highpass.

11.5 Tracking-filter asymptotic error

For reference $f(t)$, output $y(t) = T(s)F(s)$:

$$e_f = \lim_{t \rightarrow \infty} (y(t) - f(t)) = \lim_{s \rightarrow 0} s [T(s) - 1] F(s)$$

if the final-value theorem is valid.

11.6 Exam recipe: zero steady-state error for step and ramp

Use when: tracking filter $T(s) = \frac{as+b}{s^2+c_1s+c_0}$ must track step/ramp.

1. Step error zero requires $T(0) = 1$.
2. Ramp error zero additionally requires the first-order coefficient of $T(s)$ near $s = 0$ to match $1 + O(s^2)$.
3. Equivalently solve $\lim_{s \rightarrow 0} [T(s) - 1] = 0$ and $\lim_{s \rightarrow 0} \frac{T(s)-1}{s} = 0$.
4. Check stability of denominator before using final-value theorem.

Example pattern: $\frac{as+b}{(s+1)(s+2)}$ needs $b = 2$ and $a = 3$.

12 Butterworth, Sallen-Key, sensitivitet

12.1 Ctrl+F keywords

Butterworth, Sallen-Key, Sallen Key, sensitivitet, sensitivity, Monte Carlo, component tolerance, frequency scaling, impedance scaling, lowpass prototype.

12.2 Butterworth prototypes

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}.$$

For normalized $\omega_c = 1$, use stable left-half-plane poles of $1 + (-s^2)^n = 0$.

$$\text{2nd order Butterworth: } s^2 + \sqrt{2}s + 1, \quad \zeta = \frac{1}{\sqrt{2}}, \quad Q = \frac{1}{\sqrt{2}}.$$

Common normalized denominators:

$$\begin{aligned} n = 1: & s + 1, \quad n = 2: s^2 + 1.414s + 1, \\ n = 3: & (s + 1)(s^2 + s + 1), \quad n = 4: (s^2 + 0.765s + 1)(s^2 + 1.848s + 1). \end{aligned}$$

12.3 Sallen-Key lowpass

Unity-gain Sallen-Key denominator is commonly matched to

$$H(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

For equal $R_1 = R_2 = R$, $C_1 = C_2 = C$, unity gain:

$$\omega_n = \frac{1}{RC}, \quad \zeta = 1.$$

For general Sallen-Key, derive ω_n and ζ by matching the actual denominator to $s^2 + 2\zeta\omega_n s + \omega_n^2$.

12.4 Exam recipe: Sallen-Key node equations

Use when: asked whether node equations at a, b in a Sallen-Key circuit are correct.

1. Write KCL at each capacitor/resistor node using currents leaving the node.
2. Capacitor current in Laplace domain: $sC(V_1 - V_2)$.
3. Ideal op-amp buffer/non-inverting stage imposes its stated gain relation.
4. Check DC behavior: capacitors open at $s = 0$; high-frequency behavior from capacitor shorts.

Common traps: wrong sign for $sC(A - B)$; claiming highpass for a lowpass Sallen-Key topology.

12.5 Sensitivity, sensitivitet

$$S_x^F = \frac{x}{F} \frac{\partial F}{\partial x} = \frac{\partial \ln F}{\partial \ln x}.$$

$$F = AB \Rightarrow S_x^F = S_x^A + S_x^B, \quad F = A^n \Rightarrow S_A^F = n.$$

$$\frac{\Delta F}{F} \approx \sum_i S_{x_i}^F \frac{\Delta x_i}{x_i}.$$

12.6 Exam recipe: component sensitivity

Use when: asked for sensitivity or tolerance influence.

1. Express target F as a product/power if possible.
2. Apply $S_x^F = \partial \ln F / \partial \ln x$.
3. Combine independent small relative changes linearly for worst case.
4. For Monte Carlo, vary components within tolerance and recompute target; report spread.

12.7 Scaling

$$\text{frequency scaling: } R_{\text{new}} = \frac{R_{\text{old}}}{k_f} \text{ or } C_{\text{new}} = \frac{C_{\text{old}}}{k_f}, \quad k_f = \frac{\omega_{\text{new}}}{\omega_{\text{old}}}.$$

$$\text{impedance scaling: } R \mapsto k_z R, \quad C \mapsto C/k_z, \quad L \mapsto k_z L.$$

13 Butterworth højpas, AC coupling, instrumentation amplifier

13.1 Ctrl+F keywords

Butterworth highpass, højpas, AC-coupled, AC coupling, in-amp, instrumentation amplifier, differential gain, common-mode, CMRR, resistor mismatch.

13.2 Butterworth highpass

$$H_{HP,2}(s) = K \frac{s^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}$$

for a second-order Butterworth highpass.

$$H_{HP}(s) = H_{LP}\left(\frac{\omega_c}{s}\right).$$

Checks: $H_{HP}(0) = 0$, $H_{HP}(\infty) = K$, -3 dB at ω_c for Butterworth.

13.3 AC coupling

$$H_{AC}(s) = \frac{sRC}{1 + sRC} = \frac{s}{s + \omega_c}, \quad \omega_c = \frac{1}{RC}.$$

$$H_{\text{cascade}}(s) = \prod_i \frac{s}{s + \omega_{c,i}}.$$

Trap: cascaded identical highpass stages shift the combined -3 dB frequency; do not reuse a single-stage cutoff blindly.

13.4 Instrumentation amplifier

$$v_d = v_+ - v_-, \quad v_{cm} = \frac{v_+ + v_-}{2}, \quad y = A_d v_d + A_{cm} v_{cm}.$$

$$\text{CMRR} = \left| \frac{A_d}{A_{cm}} \right|, \quad \text{CMRR}_{dB} = 20 \log_{10} \left| \frac{A_d}{A_{cm}} \right|.$$

Three-op-amp in-amp gain pattern:

$$A_d = \left(1 + \frac{2R}{R_G}\right) \frac{R_2}{R_1}$$

for matched resistor pairs in the standard topology.

13.5 Exam recipe: AC-coupled in-amp checks

Use when: asked about highpass AC coupling, common-mode rejection, or resistor mismatch.

1. Split input into v_d and v_{cm} .
2. Apply AC-coupling highpass factors to the signal path.
3. Use resistor matching to decide whether common-mode ideally cancels.
4. Compute CMRR from differential and common-mode gains.

Fast checks: ideal matched subtractor has $A_{cm} = 0$; mismatch creates common-mode leakage.

14 Exam tactics and edge cases

14.1 Ctrl+F keywords

exam strategy, one-best-answer, false statement, ikke korrekt, more than one true, beginning value theorem, final value theorem, imaginary-axis poles, marginal stability.

14.2 One-best-answer and statement logic

1. If an answer option contains several claims, every claim must be true.
2. In “ikke korrekt” questions, select the false option after ordinary truth checking.
3. If numerical answers differ only by minor rounding, do not reject solely on last digits.
4. Use fast eliminators first: units, signs, $H(0)$, $H(\infty)$, pole locations, and causality.

14.3 Imaginary-axis poles and frequency response

$$H(s) = \frac{1}{s^2 + \omega_0^2} \Rightarrow p = \pm j\omega_0.$$

Ordinary $H(j\omega)$ is singular at $\omega = \omega_0$; exam options may use distribution terms for causal sinusoids.

$$\mathcal{F}\{\sin(\omega_0 t)u(t)\} = \text{PV} \frac{\omega_0}{\omega_0^2 - \omega^2} + \frac{\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

Trap: poles on the imaginary axis are not BIBO stable even if not exponentially growing.

14.4 Checklist before selecting an answer

System equation	linear? constant coefficients? causal input only?
Transfer function	zero initial conditions only? numerator/denominator powers correct?
Impulse response	direct $\delta(t)$ term included after long division?
Step/ramp response	multiply by $1/s$ or $1/s^2$, not by s .
Bode/filter type	compare $H(0)$ and $H(\infty)$.
Final value	poles of $sX(s)$ in left half-plane?